



Role of Institutional Quality in Moderating the Nexus Between Remittance Inflows and Renewable Energy Transition in the MENA Region: A Spatial Analysis

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Abstract: Remittance inflows can provide funds for the renewable energy transition (RET) in developing economies. This research aims to examine the effect of remittance inflows on RET in 11 Middle East and North Africa (MENA) economies over the period 2002–2023, while controlling for foreign direct investments (FDI), economic growth, and trade openness (TO) in the model. The moderating effect of institutional quality (IQ) is also tested in these relationships. In addition, spatial econometrics is applied due to the geographic and economic linkages among MENA economies. The results show that remittances and economic growth increase RET, with both direct localized effects and spillovers. So, these factors help raise RET in local economies as well as in their neighboring MENA economies. Governance also plays a positive moderating role in improving the effects of remittances and economic growth on RET. TO increases RET in local economies. However, the spillover effects of TO reduce RET in neighboring economies. Lastly, FDI has a statistically insignificant effect on RET in all analyses. This study recommends promoting remittance inflows and further improving IQ in the MENA region to encourage RET.

Keywords: Renewable energy transition; Remittance inflow; Spatial analysis; Economic growth; Governance; Trade openness

1 Introduction

The Middle East and North Africa (MENA) region consists of high, middle, and low-income economies. This diversity promotes inter-regional labor mobility and remittance flows [1]. The Gulf Cooperation Council (GCC) in particular is a labor-importing hub, attracting millions of laborers from other lower-income MENA economies. Egypt and Morocco, for instance, are among the top global remittance-receiving economies [2]. These remittance inflows serve as external finance for several MENA economies and can support the renewable energy transition (RET). The MENA region is trying to accelerate RET to meet global decarbonization targets. A successful RET would also reduce MENA's dependence on fossil fuels [3]. Some GCC countries have started investing in solar parks and energy efficiency initiatives [4]. However, RET remains weak in other lower-income MENA economies due to regulatory shortcomings and low renewable investments. Still, remittance inflows in these economies may support RET, which could provide a greener corridor for the whole MENA region.

Recent literature treats remittance inflows as a major driver of RET [5]. Remittances can improve the demand-side effect of RET. Workers in high-income countries may develop greater awareness of global sustainability trends. That awareness can foster RET in their home countries as well. But the effect of remittance inflows is not always positive. Remittances may raise household income and energy consumption without encouraging renewable adoption. That outcome would end up increasing fossil fuel consumption in recipient countries. Strong governance and clean-energy incentives are therefore essential. They help ensure that remittances actually have a positive impact on RET.

Strong institutions can redirect remittances toward renewable energy adoption. Transparent energy markets with strong governance may encourage households and firms to invest in renewable technologies rather than rely on cheap fossil-fuel alternatives [6]. Governance can also help integrate remittances into formal financial systems. That integration allows governments to promote green bonds, microfinance schemes, and targeted subsidies for

RET. Weak governance, however, reduces investment in renewable energy markets. It also diminishes the expected environmental benefits from remittance inflows. For these reasons, governance is treated as a moderating variable in the remittance–RET relationship.

Environmental and energy policies do not operate in isolation. They can spread across borders through geographic proximity and economic integration [7]. The MENA region likely experiences strong spatial spillovers since its countries share similar climate conditions, regional energy interdependence, and several cooperation agreements. A successful renewable energy policy in one MENA economy can benefit its neighbors and lead to cross-border transmission of energy technologies. Energy shocks or policy changes in one economy can also affect the energy strategies of neighboring economies. Remittance flows themselves have spatial patterns due to common culture, language, and labor agreements across the MENA region. Inter-regional remittance flows can therefore influence renewable energy adoption in nearby countries. These interconnections in energy and labor markets highlight why it is important to study spatial spillovers.

Existing MENA literature has investigated the nexus between energy security risks and remittance inflows [8], as well as the link between remittances and carbon emissions [9]. The present study makes several new contributions. The study tests how institutional quality (IQ) moderates the effects of economic growth, trade openness (TO), foreign direct investment (FDI), and remittance inflows on RET in a spatial analysis. This approach captures both direct effects within a country and indirect spillover effects from neighboring countries. Spatial analysis is theoretically relevant here because MENA economies share geographic proximity, energy markets, and cross-border labor flows. Therefore, this study aims to investigate the localized and spillover effects of remittance inflows on RET in the spatially linked MENA region. The findings can inform integrated policy frameworks by estimating cross-border spillovers of TO, FDI, governance, and remittance inflows on RET in the geographically connected MENA economies.

2 Literature Review

The recent literature recognizes that remittances influence energy consumption patterns and sustainable development in developing economies. Thus, remittance inflows in these economies may affect household behavior, investment capacity, and long-term energy choices.

2.1 Remittances and Energy Consumption Patterns

The recent literature has investigated the effects of remittances on energy variables in developing economies. Farzana et al. [10] examined Southeast Asia from 1991 to 2020. They reported that remittances enhanced renewable energy consumption (REC) in Bangladesh and Pakistan. However, remittances reduced REC in India and Sri Lanka. Moreover, financial deepening negatively affected REC, while urbanization and TO promoted it. Zhao and Qamruzzaman [11] investigated 59 Belt and Road Initiative (BRI) nations from 2004 to 2020. The authors showed that remittances and urbanization raised both REC and non-REC. However, globalization reduced non-REC and increased REC.

In the country-specific studies, Das et al. [12] probed Bangladesh from 1980 to 2017. They found a long-run unidirectional causality from remittances to REC in Bangladesh and a short-run feedback effect. Chakraborty [13] showed that remittance shocks had asymmetric effects on REC in India by using the period from 1990 to 2020. Negative shocks increased short-run REC and reduced long-run REC. This finding emphasized the role of institutional guidance to divert remittance inflows into REC. Hung [14] analyzed Vietnam from 1990 to 2020. Remittances supported REC in the long run but increased emissions in the short run. Seury et al. [15] examined Jamaica and found that remittances reduced REC. This corroborated a short-term tendency toward non-REC. Therefore, households prioritized energy needs over clean alternatives.

This subsection shows that earlier multi-country studies corroborate that remittances affect REC differently across countries and regions [10, 11]. These differences are due to local conditions. Moreover, urbanization, globalization, and financial development also shape these outcomes. Subsequently, some country-level studies contribute to the literature by finding the short-run versus long-run differences, asymmetric effects of remittance shocks, and feedback relationships [12–15]. A consistent insight in the literature is that remittances do not exert a uniform effect on RET. Thus, the effect of remittances is contingent on domestic absorptive capacity, policy direction, and institutional arrangements. This heterogeneity particularly highlights the importance of the role of IQ in explaining the remittance–RET relationship.

2.2 Remittances and the Environment Nexus

Panel studies explored the environmental effects of remittances. Rani et al. [16] explored South Asian Association for Regional Cooperation (SAARC) nations from 1990 to 2020. They reported that remittance inflows and energy consumption increased CO₂ emissions. Fossil fuel usage and population also increased emissions. However, the Environmental Kuznets Curve (EKC) hypothesis was supported. Ali et al. [17] confirmed asymmetric effects of remittance shocks on CO₂ emissions. Positive remittance shocks increased CO₂ emissions, and negative shocks

reduced them. However, positive shocks showed stronger effects than negative shocks. Without strategic policy measures, remittances contributed to environmental deterioration. Yang et al. [18] analyzed 97 countries from 1990 to 2016 and found that remittances increased CO₂ emissions. However, globalization reduced them.

In a cross-country analysis, Khatri et al. [19] found that remittances could not directly affect emissions. However, the positive environmental effects of remittances were realized through urbanization and TO. Thus, remittances raised environmental improvements by being strategically directed toward green investment. Moreover, the EKC hypothesis was also supported. Jamil et al. [20] explored the G-20 from 1990 to 2019 and reported that remittances significantly increased CO₂ emissions, while REC mitigated them. Gross domestic product (GDP) growth and financial development (FD) also increased emissions. Zafar et al. [21] explored 22 countries from 1986 to 2017. Remittances, export diversification, and REC decreased CO₂ emissions. Nevertheless, GDP growth and education increased them.

Majekodunmi et al. [22] analyzed D-8 countries from 1989 to 2019. Remittances mitigated CO₂ emissions by supporting green consumption. Ma and Wang [23] provided global evidence that remittances, REC, and TO enhanced long-term sustainable development. This effect mitigated the adverse impacts of natural resource exploitation. Mills [24] investigated green remittances, which combined remittances with REC, energy-efficient appliances, and climate-resilient infrastructure. However, green remittances could not affect the environment due to high transaction costs, lack of awareness, and limited institutional coordination.

Several studies focused specifically on remittance-dependent economies. Uche [25] analyzed 21 such countries from 1990 to 2019. Remittances increased emissions, but this effect became insignificant when controlling for REC, human capital, structural transformation, and urbanization. These factors acted as complementary moderators. Zhang et al. [26] studied top remittance-receiving countries. Remittances and non-REC raised ecological footprints, while REC reduced them. The EKC and pollution haven hypotheses were supported. Umair et al. [27] investigated the top 50 remittance-receiving economies from 1991 to 2018. Remittances reduced CO₂ emissions and ecological footprints. Dilanchiev et al. [28] probed top recipient economies and found an inverted U-shaped relationship between remittances and CO₂ emissions. At lower development stages, remittances increased emissions. After a threshold point, remittances reduced emissions.

Country-specific analyses added further evidence. Ali and Ali [29] analyzed Bangladesh from 1990 to 2021. Negative shocks in remittances reduced ecological footprints, but positive shocks had no effect. REC and urbanization reduced footprints, while economic growth increased them. Islam et al. [30] studied Saudi Arabia from 1990 to 2020. Remittance outflows, REC, and environmental innovation reduced CO₂ emissions. However, fossil fuel consumption led to degradation. Qiao et al. [31] analyzed India from 1975 to 2021. They found an inverted U-shaped relationship between remittance inflows and carbon intensity. Early-stage remittances increased emissions. Nevertheless, long-term remittances improved the environment through household adoption of cleaner technologies.

Abdul et al. [32] investigated China from 1990 to 2020. Remittances increased carbon intensity, reflecting carbon-intensive household consumption. Saliba et al. [33] re-examined China from 1990 to 2019. Remittances, REC, and globalization reduced CO₂ emissions, while GDP growth increased them. Khan et al. [34] found that remittance inflows increased CO₂ emissions in Australia, Germany, and India but had no effect in Mexico. This finding reflected differences in industrial structure. REC mitigated emissions, while economic growth aggravated them.

This subsection shows that earlier studies often report the pollution-enhancing effect of remittances due to fossil fuel consumption in panel analysis [16, 18, 20]. In a methodological innovation, Ali et al. [17] corroborate the asymmetrical effects. Later literature finds that remittances can improve environmental quality with moderating effects of REC, TO, and green investment [19, 21–23]. The concept of green remittances highlights the role of financial innovation and institutional coordination. Studies on remittance-dependent and top-recipient economies identify the threshold point of environmentally friendly remittances. This stream of literature also highlights the moderating role of structural and human capital factors in achieving environmental sustainability [25–28]. Country-specific studies also find nonlinear effects and asymmetric responses due to energy structure, innovation capacity, and institutional settings [29–34]. Most literature reports adverse environmental effects of remittances. However, a consistent insight across these strands is that the environmental effects of remittances depend on energy composition, economic development stages, policy environments, REC, IQ, and structural transformation.

2.3 The Moderating Role of Institutions in the Nexus between Remittances, Energy Consumption, and the Environment

Some literature also investigated the role of institutions as a moderator in the association between remittances, energy consumption, and the environment. Karmaker et al. [35] investigated this role in the nexus between REC and remittances. The authors found that remittances enhanced REC, and policy incentives supported this effect. Government subsidies and low tariffs for renewable energy products enhanced REC through remittance inflows. Yakubu et al. [36] explored 45 African economies from 2000 to 2020. Remittances reduced REC in a direct effect, but IQ moderated this relationship. The combined effect of IQ and remittances increased REC. Thus, IQ redirected

remittances toward clean energy adoption.

Qamruzzaman [37] explored the Bay of Bengal Initiative for Multi-Sectoral Technical and Economic Cooperation (BIMSTEC) economies. Remittance inflows increased the REC. Moreover, external debt, TO, and political stability also positively moderated the effect of remittances on REC. Simplice et al. [38] examined 47 African countries from 1990 to 2021. IQ moderated the effect of remittances. Remittances enhanced sustainable development by improving education and access to cleaner household energy. Zhang and Zhao [39] examined Brazil, Russia, India, and China. Remittances reduced carbon emissions. Moreover, policy uncertainty and natural resources heightened environmental degradation. However, technological innovation positively moderated these effects. Using global data, Rahaman and Islam [40] demonstrated that remittances, REC, and FD collectively reduced CO₂ emissions in an EKC framework. Thus, remittances reduced emissions through clean energy adoption in the presence of robust financial systems.

Recent empirical studies reported the heterogeneous effect of IQ in shaping the RET. Makun [41] investigated 91 economies from 2002 to 2022. Technological innovation and formal institutions promoted RET across the whole sample. In the subsample analyses, the authors found that technological innovation positively influenced RET in developed economies with a high level of IQ. The opposite finding was reported in developing economies with lower IQ. This asymmetry was corroborated because IQ moderated differently at different development levels. Similarly, Jiancheng et al. [42] found that IQ positively moderated the effect of domestic credit for renewable technologies in N-11 countries over the period 2000 to 2024. However, IQ interacted with market capitalization had a declining moderating effect on solar energy. Yet this interaction positively moderated the effect on wind and hydro technologies. Iormom et al. [43] examined Nigeria. Strong institutions and sustainable policies consistently enhanced RET, but economic policy uncertainty produced short-run fluctuations in this transition. Rao et al. [44] analyzed 45 developing countries from 2000 to 2020. REC reduced economic growth. Stronger control of corruption and the rule of law mitigated these negative effects. However, transparency and internalizing transition costs enhanced these effects. Munir [45] investigated 15 major economies from 1999 to 2023. IQ enhanced RET, and the interaction of IQ and digital transformation further promoted RET.

This subsection shows that IQ serves as a positive moderator to determine the effect of remittances on REC and environmental quality. Earlier studies find that remittances have mixed effects on REC. Remittances combined with strong IQ and technological innovation improve environmental quality [35–40]. Recent studies report heterogeneity and asymmetry in IQ's moderating role across development stages. Strong institutions and technological advancement jointly promote RET in developed countries but reduce it in developing countries [41]. Jiancheng et al. [42] report different effects of IQ on different renewable energy sources. Some studies also find consistently positive moderating effects of IQ on RET [43, 45].

The literature as a whole shows that remittances can affect sustainable energy and environmental outcomes either positively or negatively. With strong governance and policy support, remittances enhance REC adoption and reduce emissions. With weak institutions and heavy reliance on fossil fuels, remittances increase environmental problems. IQ, human development, TO, and technological innovation all help achieve the positive environmental effects of remittances. However, the literature lacks spatial analysis of the remittances–RET nexus. This gap matters particularly for geographically close MENA economies. The present research fills that gap by applying spatial econometrics to 11 MENA economies. The moderating role of governance is also tested while controlling for FDI, TO, and economic growth.

3 Methodology

3.1 Conceptual Framework, the Model, and the Data

Theoretically, remittance inflows may support RET through both microeconomic and macroeconomic channels. In the New Economics of Labor Migration theory, remittance inflows are a source of income diversification and risk sharing [46]. This income effect can increase household liquidity and investment capacity for RET. Similarly, the Energy Ladder Hypothesis suggests that increasing household income may shift traditional biomass consumption toward cleaner and modern energy sources [47]. For instance, remittances may increase household disposable income. This rising income may reduce liquidity constraints of low and middle-income households and increase RET. Remittances can support capital investment in buying solar panels, clean cooking fuels, solar water heaters, and energy-efficient appliances at the household level. Thus, remittances can reduce initial cost barriers to RET. In financial development theory, remittances may improve savings and creditworthiness, which enhance household access to green financing products through formal financial systems. In addition, migrants in high-income countries may also transfer knowledge, skills, and environmental awareness to their home countries. Those spillovers can enhance household preferences for cleaner energy choices [48].

Like the micro level, remittances also play their role through macroeconomic channels. Remittances support financial sector development by increasing the volume of deposits, the customer base for financial institutions, and liquidity in credit markets [49]. These effects can support green finance and renewable energy investments nationally. Moreover, remittances also help in transitioning energy-intensive sectors toward sustainable and service-oriented

sectors. This shift can increase the share of renewable energy at the national level. Additionally, remittances can help stabilize the balance of payments and exchange rates. These stabilizing effects can enhance aggregate investment and government capacity to invest in renewable energy infrastructure like solar parks and grid modernization [40]. In addition, remittances also improve fiscal space for governments to subsidize renewable projects [50]. On the whole, remittances can improve the standard of living and awareness of global sustainability trends. These effects can put demand-side pressure on renewable energy policies.

IQ can also shape the effect of remittances on RET. For instance, institutions can create economic incentives and supportive policies [51, 52]. They reduce uncertainty and transaction costs for households and large-scale renewable energy investments. Strong institutions may enhance transparency, accountability, and policy credibility [53]. These features can encourage households and firms to allocate funds to long-term renewable energy investments. Financial systems with strong regulatory support may also transform remittances into savings, credit, and investment capital for renewable technologies [54]. Moreover, good governance may facilitate technological diffusion by supporting innovation systems and protecting property rights [49].

Based on these arguments, the present study hypothesized the positive moderating effect of governance on the nexus between remittance inflows and RET. Additionally, FDI and TO might also serve as foreign assistance for RET. Therefore, FDI, TO, and economic growth were also added to the following model to examine the moderating effect of governance:

$$RET_{it} = f(Y_{it}, REM_{it}, FDI_{it}, TO_{it}, GOV_{it}, MOD_{it}) \quad (1)$$

The proposed model is tested for a panel of 11 MENA economies (i), including Algeria, Egypt, Iran, Iraq, Israel, Kuwait, Morocco, Oman, Qatar, Saudi Arabia, and the United Arab Emirates (UAE). t shows the period from 2002 to 2023. The time and countries' samples are chosen as per data availability. The cross-sectional dimension ($i = 11$) is modest, and the time sample (t) is also relatively limited. However, the spatial panel models can be reliably estimated with ($i = 5$), when the time sample is sufficiently large ($t > 20$) [55]. Therefore, the estimated panel model is adequately reliable, though not very strong for stability concerns.

RET_{it} represents the renewable energy transition, defined with a REC in exajoules. The data on RET is sourced from [56]. Conceptually, RET captures structural changes in energy systems through policy frameworks, infrastructure development, and institutional reforms. However, consumption-based measurement of renewable energy represents both supply-side capacity and demand-side adoption. This provides a comprehensive indicator of the transition progress of renewable energy. Moreover, the choice of absolute consumption in exajoules provides consistent cross-country data, which is essential for spatial econometric analysis. The recent literature has also utilized REC as a proxy of RET [41, 42].

Y_{it} is the natural log of real GDP per capita in constant USD, REM_{it} captures remittance inflows percentage of GDP. FDI_{it} denotes FDI net inflow as a percentage of GDP. TO_{it} is the total trade percentage of GDP. GOV_{it} is a governance index. All macroeconomic series, except RET, are sourced from study [57]. MOD_{it} is introduced in the model to capture the moderating effect of governance on the relationship between all independent variables and RET.

Some series were interpolated to complete the missing data. For this purpose, linear interpolation was applied to series with occasional gaps, as economic variables usually exhibit relatively smooth trends over time. This method estimates missing values by connecting known data points with a straight line, assuming a constant rate of change between observations. Moreover, the missing observations were less than 5% of the total dataset, which minimized the risk of introducing bias into the dataset [58]. In addition, all interpolated series were visually inspected to ensure that imputed values aligned with the underlying trend.

3.2 Spatial Methodology

The MENA region consists of geographically closed economies and shares common environmental and economic features. These conditions suggest the expected spatial interconnectedness. This region in particular has a common energy mix, which can create spatial linkages in RET. Moreover, remittance inflows could also have spatial linkages because of cross-border labor flows within the MENA region. These labor flows can create RET spillovers from one MENA economy to other economies. Furthermore, MENA countries also maintain strong trade, FDI, and policy linkages. These linkages can enhance the potential for spatial dependence in governance, RET, trade, and investment flows. All of these factors can affect energy consumption patterns and policies across the MENA region.

Given the expected spatial interdependence in MENA economies, it looks pertinent to examine spatial autocorrelation. The Lagrange Multiplier (LM) and robust LM have strong power to detect spatial autocorrelation in such models [59]. Both tests can capture omitted spatially lagged variables to reduce bias in regression estimates. Moreover, Moran's I measures the expected spatial clustering in the dependent variable [60]. In the presence of statistically significant spatial autocorrelation, ignoring spatial effects could lead to biased regression results. Therefore, the Spatial Durbin Model (SDM) was utilized to capture both direct effects within a country and indirect

spillover effects from neighboring countries:

$$\begin{aligned}
RET_{it} = & \beta_0 + \beta_{11}Y_{it} + \beta_{12}REM_{it} + \beta_{13}GOV_{it} + \beta_{14}TO_{it} + \beta_{15}FDI_{it} + \beta_{16}MOD_{it} \\
& + \beta_{21} \sum_{j=1}^n w_{ij}Y_{it} + \beta_{22} \sum_{j=1}^n w_{ij}REM_{it} + \beta_{23} \sum_{j=1}^n w_{ij}GOV_{it} \\
& + \beta_{24} \sum_{j=1}^n w_{ij}TO_{it} + \beta_{25} \sum_{j=1}^n w_{ij}FDI_{it} + \beta_{26} \sum_{j=1}^n w_{ij}MOD_{it} \\
& + \delta_1 \sum_{j=1}^n w_{ij}RET_{it} + v_i + u_t + e_{it}
\end{aligned} \tag{2}$$

In Eq. (2), non-weighted variables detected the direct effect within a local economy. Weighted variables captured the spillovers from one MENA economy to other neighboring economies. w_{ij} is a weight matrix. This matrix was constructed using the inverse distance between MENA economies. A higher w_{ij} showed the closeness of MENA economies.

$$w_{ij} = \begin{cases} \frac{1}{d_{ij}}, & \text{for } i \neq j \\ 0, & \text{for } i = j \end{cases} \tag{3}$$

In Eq. (3), d_{ij} represents the distance between MENA economies. The distance was taken as inverse to capture the closeness of economies. The distance was measured as the great-circle distance (in kilometers) between the capital cities of MENA countries using an online distance calculator (<https://www.distance.to/>). This provided the shortest geographical distance between two countries based on the capital city of each country, which ensured consistency and comparability across countries. The distance matrix was preferred over the contiguity matrix. Because the contiguity matrix captured binary numbers and overlooked geographic proximity in non-contiguous countries. Moreover, geographical distance was exogenous and time-invariant. In another weight matrix approach, the economic distance matrix introduced endogeneity because the model already included TO, economic growth, and FDI. Therefore, the study chose the inverse-distance spatial weight matrix to estimate spillovers. To estimate consistent spillovers, w_{ij} was row-standardized by following the methodology of Kelejian and Prucha [61].

All spatial procedures were estimated with Stata software. Moreover, other spatial specifications, including the Spatial Autoregressive (SAR), Spatial Autocorrelation (SAC), and Spatial Error Model (SEM), were also estimated. This approach helped to find the most suitable spatial specification for the proposed model. For this purpose, Akaike information criterion (AIC) and Bayesian information criterion (BIC) were applied to all estimated spatial specifications. In addition, spatial error and lag tests were also used to compare the SDM over SAR and SEM.

4 Results and Discussions

In Table 1, the descriptive statistics show mild variations in the variables. RET shows a low average value, which indicates a slow progress of REC in the MENA economies. Economic growth (Y_{it}) is relatively stable across MENA countries due to their common economic landscape. However, remittance inflows (REM_{it}) and trade openness (TO_{it}) show more fluctuations, showing structural differences among MENA economies. Governance (GOV_{it}) centers on zero with a heavy variation. This variation reflects a mixed institutional performance across MENA economies. FDI inflows (FDI_{it}) also vary significantly, which reflects dynamic variations in FDI inflows in the MENA economies.

Table 1. Descriptive statistics

Variables	Observations	Mean	S.D.	Minimum	Maximum
RET_{it}	242	0.0167	0.0235	0	0.1264
Y_{it}	242	9.4359	1.1413	7.6627	11.3097
REM_{it}	242	1.3095	2.4512	0.0001	9.9602
TO_{it}	242	78.4458	32.6644	0.0001	202.3325
GOV_{it}	242	0.0256	0.7751	-1.9626	1.6043
FDI_{it}	242	1.8381	2.2925	-4.5416	11.7936

Table 2 shows the correlation matrix. RET correlates positively with most variables. Economic growth, remittance inflows, and governance may therefore support RET in the MENA region. Economic growth shows the strongest correlation with RET. FDI inflows correlate negatively. This relationship suggests that foreign investment flows toward fossil fuel-dependent sectors rather than supporting RET. Most correlations are moderate, indicating a low chance of multicollinearity.

Table 2. Correlation matrix

Variable	RET_{it}	Y_{it}	REM_{it}	TO_{it}	GOV_{it}	FDI_{it}
RET_{it}	1					
Y_{it}	0.510	1				
REM_{it}	0.462	0.199	1			
TO_{it}	0.365	0.129	-0.305	1		
GOV_{it}	0.384	0.364	0.240	0.437	1	
FDI_{it}	-0.159	0.178	0.110	0.235	0.157	1

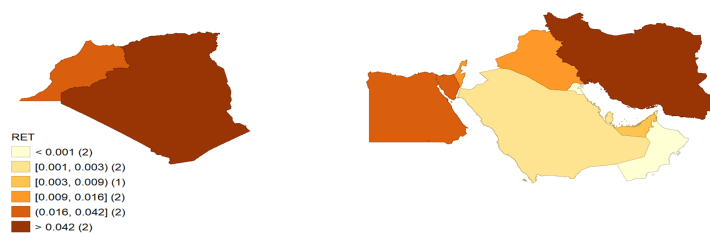
Tables 3–5 present the main results. Models 1 through 4 are estimated separately to capture the moderating role of governance on the nexus between RET and REM_{it} , Y_{it} , TO_{it} , and FDI_{it} , respectively. In Table 3, the non-spatial regressions are estimated. The results show that Y_{it} , REM_{it} , TO_{it} , and GOV_{it} have positive effects on the RET. However, FDI_{it} shows no significant effect. Moreover, GOV_{it} positively moderates all relationships except for FDI_{it} . However, Moran's I and LM statistics confirm spatial autocorrelation in all models. The results in Table 3 are therefore not reliable for interpretation due to omitted spatial dependence. These are only presented to demonstrate why spatial econometrics is methodologically necessary here.

Table 3. Non-spatial models

Variables	Model 1	Model 2	Model 3	Model 4
Y_{it}	0.028 (0.006)	0.031 (0.006)	0.029 (0.002)	0.026 (0.008)
REM_{it}	0.027 (0.001)	0.009 (0.002)	0.006 (0.011)	0.008 (0.016)
TO_{it}	0.009 (0.000)	0.002 (0.000)	0.001 (0.001)	0.002 (0.000)
GOV_{it}	0.049 (0.003)	0.071 (0.028)	0.001 (0.005)	0.024 (0.003)
FDI_{it}	-0.001 (0.406)	-0.016 (0.403)	-0.003 (0.411)	-0.001 (0.415)
$REM_{it} * GOV_{it}$	0.031 (0.002)	–	–	–
$Y_{it} * GOV_{it}$	–	0.073 (0.028)	–	–
$TO_{it} * GOV_{it}$	–	–	0.001 (0.012)	–
$FDI_{it} * GOV_{it}$	–	–	–	-0.001 (0.527)
Intercept	-0.264 (0.063)	-0.294 (0.066)	-0.271 (0.064)	-0.248 (0.065)
Diagnostic tests				
R^2	0.527	0.318	0.789	0.663
F-value	52.134 (0.000)	12.847 (0.000)	98.657 (0.000)	48.215 (0.000)
Hausman test	131.742 (0.000)	89.315 (0.000)	261.874 (0.000)	145.692 (0.000)
Moran's I	16.248 (0.000)	4.712 (0.000)	9.037 (0.000)	6.213 (0.000)
LM-error	187.903 (0.000)	20.836 (0.000)	50.912 (0.000)	24.105 (0.000)
LM Robust-error	5.827 (0.018)	6.742 (0.011)	11.637 (0.001)	8.103 (0.005)
LM-lag	278.452 (0.000)	23.714 (0.000)	70.284 (0.000)	18.639 (0.000)
LM Robust-lag	105.937 (0.000)	19.836 (0.000)	32.118 (0.000)	12.745 (0.000)

Note: p -values are presented in parentheses. A dash (–) denotes no data. LM, Lagrange Multiplier.

Figure 1 shows the spatial distribution of RET based on average values from 2002 to 2023. Algeria shows the highest level of RET, consistent with its early investments in solar energy and large-scale renewable development plans. North African countries show better RET compared to Middle Eastern countries. North African countries historically pursued renewable energy development as a strategy to reduce fossil fuel dependence. This effort enhanced energy security in their countries. In the Middle East region, Iran shows the highest level of RET. This finding reflects the hydropower capacity of Iran and other efforts to diversify energy sources. Moreover, Israel, the UAE, and Qatar also carry a significant RET, reflecting their significant investment in solar power plants. Figure 1 reveals spatial linkages in RET among sample countries. Geographically proximate North African countries show relatively higher RET levels. Similarly, proximate GCC countries show comparable RET levels.

**Figure 1.** Spatial distribution of renewable energy transition (RET)

Note: This figure was prepared by the author, and the map is based on the average values of RET during 2002–2023.

Figure 2 shows the spatial distribution of remittance inflows. Israel ranks first due to its large and wealthy diaspora in the United States and France. These migrants maintain strong financial, cultural, and religious connections with their home country. They send remittances for investment, charitable contributions, and real estate. The UAE, Oman, Egypt, and Morocco also attract significant remittances. Egypt and Morocco are among the world's top remittance-receiving countries. Their large migrant populations work in Gulf countries, Europe, and North America. Remittance inflows in the UAE and Oman are high due to expatriate financial activities and international transactions. Figure 2 shows spatial linkages in remittance inflows across selected MENA economies. Geographically proximate Egypt, Morocco, and Algeria show relatively higher remittance levels. The UAE and Oman are also geographically close. A comparison of Figure 1 and Figure 2 shows that Algeria, Egypt, Morocco, Israel, and the UAE have relatively higher levels of both remittance inflows and RET. This pattern suggests potential spatial linkages between remittances and RET that warrant further spatial investigation.

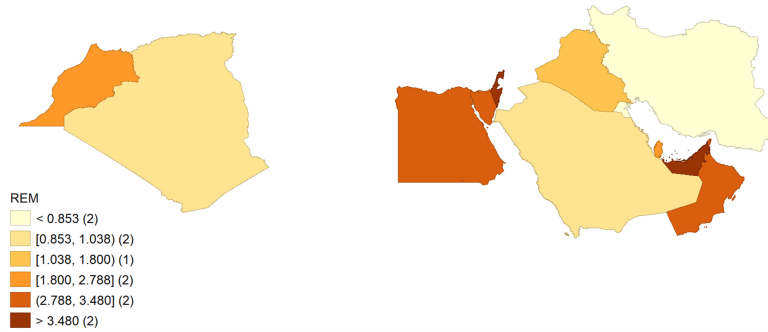


Figure 2. Spatial distribution of remittance inflows

Note: This figure was prepared by the author, and the map is based on the average values of remittance inflows during 2002–2023.

After confirming spatial autocorrelation, AIC, BIC, and Hausman tests were applied to identify the most appropriate spatial specification. In Table 4, the Hausman test consistently shows that fixed effects are preferable to random effects across SDM, SAC, SAR, and SEM specifications. AIC and BIC values are lowest for the SDM-FE specification. This confirms that SDM-FE is the most appropriate spatial model for analyzing RET in the MENA region with our sample.

Table 4. Spatial specification selection

Tests	Model 1	Model 2	Model 3	Model 4
SDM FE-AIC	-159.237	612.894	53.782	1308.621
SDM FE-BIC	-112.512	575.432	88.346	1224.739
Hausman test	249.873 (0.000)	121.567 (0.000)	305.982 (0.000)	146.428 (0.000)
SAC FE-AIC	-142.158	618.743	51.429	1312.894
SAC FE-BIC	-97.624	682.891	90.215	1331.217
Hausman Test	253.214 (0.000)	119.843 (0.000)	312.765 (0.000)	150.981 (0.000)
SAR FE-AIC	-111.562	632.108	101.245	1332.417
SAR FE-BIC	-86.743	659.217	126.384	1357.528
Hausman test	202.481 (0.000)	101.376 (0.000)	134.572 (0.000)	91.547 (0.000)
SEM FE-AIC	-105.874	635.762	114.621	1334.879
SEM FE-BIC	-80.945	660.891	139.845	1359.894
Hausman test	273.582 (0.000)	155.407 (0.000)	359.241 (0.000)	220.184 (0.000)

Note: *p*-values are presented in parentheses. SDM, Spatial Durbin Model; SAC, Spatial Autocorrelation; SAR, Spatial Autoregressive Model; SEM, Spatial Error Model; FE, Fixed Effects; AIC, Akaike information criterion; BIC, Bayesian information criterion.

After validating the appropriateness of SDM-FE, the Wald and Likelihood Ratio (LR) tests were applied to all models. In Table 5, both tests show highly significant statistics, which explains that SDM-FE does not tend to be reduced to any other spatial specification. Thus, SDM-FE provides the most robust results for all RET models. The coefficient of $W*RET_{it}$ is positive and the largest among all weighted coefficients. This result explains that the RET shows strong regional spillovers among MENA economies. The coefficient of GOV_{it} is positive in direct estimates. Thus, governance helps to improve RET in local economies. However, the spillovers of GOV_{it} are insignificant. Thus, good governance in one MENA country could not influence RET in neighboring economies. Nevertheless, the total effect of governance is positive, which confirms that governance is an important factor of the RET in the whole MENA region.

Table 5. Spatial models

Regressor	Model 1	Model 2	Model 3	Model 4
		Main		
Y_{it}	0.019 (0.000)	0.021 (0.000)	0.018 (0.000)	0.018 (0.001)
REM_{it}	0.018 (0.112)	0.005 (0.571)	0.008 (0.365)	0.008 (0.355)
TO_{it}	0.002 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
GOV_{it}	0.057 (0.043)	0.055 (0.011)	0.007 (0.094)	0.049 (0.088)
FDI_{it}	-0.002 (0.695)	-0.001 (0.659)	-0.001 (0.860)	-0.001 (0.758)
$REM_{it}*GOV_{it}$	0.019 (0.020)	—	—	—
$Y_{it}*GOV_{it}$	—	0.052 (0.021)	—	—
$TO_{it}*GOV_{it}$	—	—	0.001 (0.473)	—
$FDI_{it}*GOV_{it}$	—	—	—	-0.001 (0.918)
		Direct		
Y_{it}	0.021 (0.000)	0.024 (0.000)	0.021 (0.000)	0.020 (0.000)
REM_{it}	0.029 (0.008)	0.012 (0.027)	0.018 (0.062)	0.017 (0.070)
TO_{it}	0.001 (0.000)	0.001 (0.004)	0.001 (0.004)	0.001 (0.004)
GOV_{it}	0.061 (0.033)	0.067 (0.006)	0.061 (0.023)	0.046 (0.027)
FDI_{it}	-0.001 (0.664)	-0.002 (0.597)	-0.001 (0.869)	-0.001 (0.823)
$REM_{it}*GOV_{it}$	0.032 (0.032)	—	—	—
$Y_{it}*GOV_{it}$	—	0.064 (0.011)	—	—
$TO_{it}*GOV_{it}$	—	—	0.002 (0.670)	—
$FDI_{it}*GOV_{it}$	—	—	—	-0.001 (0.911)
		Indirect		
Y_{it}	0.036 (0.061)	0.057 (0.014)	0.043 (0.087)	0.043 (0.084)
REM_{it}	0.033 (0.000)	0.016 (0.000)	0.021 (0.000)	0.019 (0.000)
TO_{it}	-0.006 (0.000)	-0.007 (0.000)	-0.007 (0.000)	-0.008 (0.000)
GOV_{it}	0.095 (0.377)	0.063 (0.151)	0.022 (0.414)	0.006 (0.671)
FDI_{it}	-0.001 (0.899)	-0.004 (0.688)	0.002 (0.865)	0.006 (0.608)
$REM_{it}*GOV_{it}$	0.035 (0.000)	—	—	—
$Y_{it}*GOV_{it}$	—	0.026 (0.048)	—	—
$TO_{it}*GOV_{it}$	—	—	-0.001 (0.466)	—
$FDI_{it}*GOV_{it}$	—	—	—	-0.008 (0.792)
		Total		
Y_{it}	0.057 (0.005)	0.081 (0.001)	0.064 (0.020)	0.063 (0.020)
REM_{it}	0.062 (0.000)	0.028 (0.000)	0.039 (0.000)	0.036 (0.000)
TO_{it}	-0.005 (0.001)	-0.006 (0.000)	-0.006 (0.001)	-0.007 (0.000)
GOV_{it}	0.156 (0.018)	0.130 (0.027)	0.083 (0.061)	0.052 (0.095)
FDI_{it}	-0.002 (0.793)	-0.006 (0.601)	0.001 (0.909)	0.005 (0.677)
$REM_{it}*GOV_{it}$	0.067 (0.000)	—	—	—
$Y_{it}*GOV_{it}$	—	0.090 (0.027)	—	—
$TO_{it}*GOV_{it}$	—	—	0.001 (0.569)	—
$FDI_{it}*GOV_{it}$	—	—	—	-0.009 (0.803)
		Weighted		
$W*Y_{it}$	0.021 (0.020)	0.029 (0.039)	0.017 (0.021)	0.018 (0.019)
$W*REM_{it}$	0.024 (0.000)	0.011 (0.001)	0.012 (0.000)	0.015 (0.000)
$W*TO_{it}$	-0.004 (0.012)	-0.005 (0.056)	-0.005 (0.033)	-0.001 (0.011)
$W*GOV_{it}$	0.053 (0.459)	0.157 (0.072)	0.015 (0.332)	0.062 (0.482)
$W*FDI_{it}$	-0.001 (0.985)	-0.002 (0.785)	0.002 (0.784)	0.005 (0.536)
$W*REM_{it}*GOV_{it}$	0.025 (0.000)	—	—	—
$W*Y_{it}*GOV_{it}$	—	0.016 (0.067)	—	—
$W*TO_{it}*GOV_{it}$	—	—	-0.001 (0.418)	—
$W*FDI_{it}*GOV_{it}$	—	—	—	-0.006 (0.706)
$W*RET_{it}$	0.281 (0.002)	0.366 (0.000)	0.435 (0.000)	0.422 (0.000)
		Diagnostics tests		
R^2	0.513	0.467	0.432	0.434
Variance	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Hausman test	241.732 (0.000)	129.452 (0.000)	287.910 (0.000)	167.204 (0.000)
Wald-spatial lag	42.735 (0.000)	28.412 (0.000)	51.925 (0.000)	103.874 (0.000)
Wald-spatial error	46.589 (0.000)	19.345 (0.000)	57.211 (0.000)	121.492 (0.000)
LR-spatial lag	34.964 (0.000)	22.175 (0.000)	43.887 (0.000)	94.724 (0.000)
LR-spatial error	32.895 (0.000)	21.633 (0.000)	29.479 (0.000)	73.581 (0.000)

Note: p -values are presented in parentheses. A dash (—) denotes no data. LR, Likelihood Ratio.

In the direct effects, Y_{it} has a positive effect on RET. Higher income levels within a country stimulate renewable energy adoption. Y_{it} also has a positive indirect effect, meaning that higher growth in one country benefits RET in

neighboring economies. The total effect of Y_{it} is positive. Therefore, economic growth supports RET across the entire MENA region. Additionally, the positive effect of $Y_{it} * GOV_{it}$ on RET_{it} in all estimates shows that governance helps translate economic progress into RET by providing coherent policy frameworks for renewable energy adoption.

REM_{it} increases RET_{it} directly. The spillover effects of remittances are also positive and even stronger than the direct effect. The total effect of REM_{it} is therefore positive. Both the direct and indirect effects of $REM_{it} * GOV_{it}$ are positive, and the total effects are also positive. Remittance inflows in the presence of strong governance help accelerate RET across the whole MENA region.

TO_{it} increases RET_{it} in direct effect. However, its indirect effect on RET_{it} is negative. The total effects of TO_{it} are also negative. The effects of $TO_{it} * GOV_{it}$ are statistically insignificant in all estimates. FDI has a statistically insignificant effect in all estimates. The effects of $FDI_{it} * GOV_{it}$ are also insignificant. Therefore, governance does not positively moderate the relationship between FDI and RET.

5 Discussions

Economic growth has a direct positive effect on RET. This finding confirms that higher income levels stimulate renewable energy adoption within MENA countries. The effect is especially visible in high-income GCC countries and some North African countries, which have launched many solar and renewable energy projects in recent years. This result aligns with the energy transition literature, which argues that economic development enables countries to allocate resources toward cleaner energy infrastructure [3]. Economic growth develops the financial capacity and technological expertise needed for RET. The spillover and total regional effects of economic growth are also positive. Economic expansion in one MENA country, therefore, fosters RET in neighboring countries through technology and knowledge diffusion from high-RET countries to low-RET neighbors.

Remittance inflows have a direct positive effect on RET in MENA economies. This finding supports our conceptual framework, which emphasizes household liquidity and investment capacity channels. At the macroeconomic level, remittance inflows create an aggregation effect. Individual household RET decisions collectively translate into improved national RET outcomes. This result is consistent with earlier studies [10, 11, 14, 35]. These studies also uniformly found that remittances promote REC in developing countries by enhancing purchasing power for clean energy technologies. Our results also show positive spillovers and total effects of remittances on RET. This finding is theoretically grounded in migration network theory. Adoption of renewable technologies in one MENA country diffuses knowledge and practices to neighboring countries through family networks, cultural connections, and shared labor markets.

IQ has direct and total positive effects on RET in MENA economies. The direct positive effect of IQ on RET is in line with recent literature [41, 43, 45]. This result corroborates that strong institutions create a supportive environment for renewable energy investment in the MENA region, accelerating RET. Governance helps reduce policy uncertainty and corruption in renewable energy markets as per theoretical expectations. Thus, governance helps in implementing renewable energy policies to attract private investment in renewable projects. However, the spillovers of governance are insignificant. Thus, governance is a more domestic phenomenon in the MENA economies. It is not being diffused to neighboring MENA economies to support RET.

TO has a direct positive effect on RET. International trade facilitates access to renewable energy technologies, equipment, and expertise. Most MENA economies have lower tariffs on imports, which may reduce the costs of importing solar panels, wind turbines, and other renewable technologies. However, the spillovers and total effects of TO on RET are negative. This finding suggests that TO increases regional competition for the energy market. That competition motivates neighboring countries to prioritize fossil fuels for a competitive advantage. Moreover, there is also intra-regional fossil fuel trade in some MENA economies through long-term supply contracts. This dynamic can create a fossil fuel lock-in effect, which discourages neighboring countries from adopting RET strategies. For instance, the GCC Interconnection Authority facilitates cross-border electricity trade among Gulf countries produced from fossil fuels. Similarly, TO may increase regional dependence on non-renewable energy sources, which undermines regional RET efforts in MENA countries.

FDI has an insignificant effect in all direct, spillover, and regional estimates. This result reflects the fact that most FDI inflows have historically flowed into fossil fuel extraction, petrochemical industries, and real estate sectors instead of flowing into renewable energy infrastructure. FDI in renewable energy requires complementary domestic capabilities of technical expertise and maintenance capacity, which are absent in most MENA economies. However, Morocco is an outlier by developing a renewable energy ecosystem, which attracts renewable energy FDI. Saudi Arabia, the UAE, and Egypt have recently launched large-scale renewable energy initiatives with the help of foreign expertise and investment. But these efforts fall outside the study's sample period (2002–2023). Lastly, fossil fuel energy subsidies in some MENA countries may also discourage renewable energy FDI in the region.

The moderating role of governance is positive and significant for the remittances-growth nexus across direct, spillover, and regional estimates. IQ enhances the effectiveness of remittances and economic growth for RET. This finding validates the theoretical argument that governance channels household-level financial inflows toward

RET. Strong governance provides regulatory support, consumer protection, and financial intermediation. These factors motivate remittance-receiving households to invest in renewable technologies. This result aligns with earlier studies [35–38]. Moreover, governance also helps translate economic progress into RET by providing effective policy frameworks and reducing barriers to clean energy investment. The spillovers of RET are positive and significant. Thus, RET in the MENA region is an integrated regional process rather than a set of isolated domestic processes. Regional energy cooperation, cross-border renewable energy projects, and policy learning among MENA countries all support this outcome.

6 Conclusions and Policy Implications

This study investigates the effect of remittance inflows on RET in the geographically and economically integrated MENA region for the period 2002 to 2023. The SDM is used to capture both local effects and cross-border spillovers. Additionally, the moderating effect of governance has been examined in these relationships. The results demonstrate that RET has strong spillover effects. Thus, rising RET in a MENA country is found to be helpful in accelerating RET in neighboring MENA countries through renewable technology diffusion and knowledge transfers. Governance raises RET in local economies, though its spillovers are statistically insignificant. The total effects of governance on RET are statistically significant. Economic growth is a major driver of RET with strong positive direct effects and spillovers. Governance positively moderates the relationship between GDP per capita and RET across direct, indirect, and total effects. Remittance inflows improve RET through both direct and spillover effects across the whole MENA region. Governance positively moderates this relationship, highlighting its role in channeling household financial inflows toward clean energy adoption. Trade openness shows positive direct effects but negative spillovers. Governance does not moderate the TO–RET relationship. FDI has no statistically significant effect on RET in any estimate, and governance does not moderate this relationship. Overall, RET in the MENA region is shaped by economic progress, remittance inflows, governance, and spatial interdependence.

The findings offer several region-specific policy implications. The spillover effects of RET suggest that renewable energy policies should be regionally coordinated through cross-border electricity trade, regional renewable energy platforms, and joint green strategies. The positive spillovers of remittances indicate that remittance inflows should be directed toward renewable energy investment and green consumption through targeted financing schemes. Subsidizing REC would also promote demand-side effects. The positive direct effect of remittances on RET suggests that policymakers should integrate remittance channels into green finance strategies. This recommendation is particularly important for top remittance-receiving North African economies. These governments should design remittance-linked renewable energy funds. Moreover, rooftop solar programs should be subsidized and financed by remittance-backed credit schemes.

The results also highlight governance as a key enabling mechanism for RET. Strengthening governance in renewable energy subsidies would accelerate the transition. The positive moderating role of governance in the remittances–RET nexus indicates that policymakers should develop strong institutional frameworks to convert household remittance inflows into productive green investments. Moreover, economic growth also promotes RET with spillovers. Thus, MENA governments should further decouple growth strategies from fossil fuel dependency. This policy is particularly important for oil-exporting GCC economies. In addition, the results of the negative spillover effects of TO suggest that trade policies should include environmental regulations to prevent pollution transfer across neighboring countries. Moreover, incentives should be provided for inter-regional clean technology trade. The insignificant effect of FDI suggests the need to attract environmentally friendly and renewable technology-oriented foreign investment.

The research faces data constraints in terms of time period and country coverage in the MENA region. However, future research may expand the scope of the study by including uncovered MENA countries. Moreover, subgroup analyses could be conducted to compare oil-exporting versus oil-importing MENA economies. Cross-regional comparisons can also be investigated with Sub-Saharan Africa or Southeast Asia. These comparisons could help identify whether the observed patterns are unique to the MENA context or have broader dynamics in other remittance-dependent developing economies.

The temporal scope can also be expanded beyond the current time sample to examine pre-2000 baseline conditions and capture long-term policy shifts in renewable transition. Thirdly, the effect of remittances can be tested on disaggregated renewable energy indicators by source (solar, wind, hydro, biomass) and by sector (residential, industrial, commercial, transport). These analyses could help understand the sectoral or energy-source effects of remittances.

Additional environmental and energy indicators, such as carbon emissions, ecological footprint, air quality measures, and energy intensity, can be explored. These analyses could help clarify whether remittance-driven RET translates into measurable environmental improvements or not. Lastly, methodological extensions can include non-geographic spatial linkages based on trade flows or technological similarity to capture the temporal evolution of spillovers.

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Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The author declares no conflicts of interest.

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